

# Empiria: A Composable, Browser-Based Environment for Simulation-Based Statistics Education

Kevin Schoenholzer

May 2026

## Abstract

Introductory statistics is often taught through static equations and definitions delivered before the underlying quantities can be manipulated. We present Empiria, a free, open-source, browser-based environment that inverts this sequence. Drawing on the older tradition of analog computing, Empiria represents a statistical analysis as a manipulable dataflow graph. Students wire small computational modules together with cables on a canvas, and on each tick of a global clock every intermediate quantity updates in real time. The formula each module computes can be displayed alongside its visualization. Because the canvas is modular and customizable, students can extend, rewire, and rebuild an analysis rather than read it off a page. Empiria requires no installation, ships sixteen guided lessons and a tour, and exports figures, data, and reproducible patches. Its closed-form routines are verified against R to machine precision, and its randomized procedures against published reference values. Empiria is available at <https://kevinschoenholzer.com/empiria/>.

Keywords: Teaching Statistics; simulation-based inference; statistics education; sampling distributions; bootstrap; reproducibility; educational technology; dataflow

## Introduction

The case for reorganizing introductory statistics around simulation-based reasoning, rather than closed-form normal-theory derivation, is by now well established. Cobb (2007) argued that the curriculum had over-invested in analytic derivations at the expense of randomization- and simulation-based approaches in which the sampling distribution is built empirically before it is named; the simulation-based-inference movement subsequently built curricula around that idea (Rossman & Chance, 2014) and reported measurable gains in students' reasoning about  $p$ -values and confidence intervals (Tintle et al., 2015), and the GAISE College Report codified the corresponding recommendations: foster active learning, use real data, integrate technology, and emphasize statistical thinking (American Statistical Association,

2016). A sustained research literature documents *why* this is hard: students struggle to form durable intuitions about sampling variability and the sampling distribution of an estimator, and they benefit from manipulating parameters and observing the consequences in real time (Chance et al., 2004; delMas et al., 2007; Garfield & Ben-Zvi, 2009).

The pedagogical principle is now mainstream; what remains uneven is the *toolkit* through which it is delivered. The existing landscape is fragmented. Sampling-distribution visualizers address sampling variability (Chance et al., 2004; delMas et al., 2007); TinkerPlots supports school-level exploratory analysis (Konold & Lehrer, 2008) and CODAP extends this to data science for younger learners (Finzer, 2013); NetLogo is the dominant agent-based-modeling environment (Wilensky, 1999); StatKey supports a full simulation-based-inference curriculum in the browser (Lock et al., 2021); and a large number of single-purpose applets (for example, the *Rossmann/Chance* collection and *Seeing Theory*) and instructor-built R Shiny dashboards address specific techniques. Each of these tools is effective within its domain, but each is also *self-contained*: the output of one cannot be piped into the next analytical step. This matters because the workflow students must internalize (data-generating process  $\rightarrow$  sampling  $\rightarrow$  estimation  $\rightarrow$  inference  $\rightarrow$  diagnostics) is exactly a *composable pipeline*, and one-applet-per-concept tools render that pipeline invisible.

This article describes Empiria, an open-source, browser-based environment that restores composition at the level of the platform. In Empiria a student builds an analysis as a *dataflow graph*: small modules (“nodes”) are connected by cables, a global clock advances the computation, and the output of any node is a signal that can be routed into any other. The entire inferential pipeline therefore sits on a single screen, wired together, with every intermediate quantity visible and manipulable as it updates. The contributions of the tool, and of this article, are: (1) a composable, signal-flow interaction model for simulation-based inference that makes the inferential workflow explicit; (2) a live mathematical-notation view that renders each module’s formula with the current values substituted, tying symbolic form to direct manipulation; (3) a zero-installation, cross-device implementation with sixteen ready-made lessons and a guided tour; and (4) a verification methodology (exact numerical recipes, deterministic seeding, and an automated test suite checked against R) that makes the tool auditable rather than merely persuasive.

## Background and related tools

Empiria’s nearest neighbors are the sampling-distribution visualizers studied in the statistics-education literature (Chance et al., 2004; delMas et al., 2007), TinkerPlots (Konold & Lehrer, 2008), CODAP (Finzer, 2013), NetLogo (Wilensky, 1999), the StatKey tools that accompany a simulation-based-inference curriculum (Lock et al., 2021), the *Rossmann/Chance* and *Seeing Theory* applet collections, and the many instructor-authored R Shiny applications. These tools established that dynamic, manipulable visualization supports the construction of sampling intuitions, and Empiria builds directly on that finding.

Empiria differs from these precedents in three respects. First, it is a *composable signal-flow environment* rather than a single-purpose application. Even free, browser-based simulation

tools such as StatKey present one analysis per screen; in Empiria the output of any module is a signal that can be routed into any other, so a parametric sampler can feed a sampling-window estimator, which can feed a hypothesis test or a bootstrap, with no copy-paste or context switch. Second, it lowers the cost of authoring new teaching material: where a Shiny applet requires fluency in R, HTML, and reactive programming, an Empiria activity is assembled by wiring existing modules, and a new module is a single function with a small parameter schema. Third, it treats numerical correctness and reproducibility as first-class, testable properties (described below), so that the tool can be audited by a skeptical instructor rather than taken on trust. Empiria does not aim to replace analytic software such as R; it is a complementary, simulation-first interface intended to make the elementary operations of sampling, estimation, inference, and diagnosis directly manipulable.

## Design

### The dataflow canvas

Empiria presents an infinite canvas on which the user places typed *nodes* and connects their ports with *cables*. Every node exposes labelled input ports on its left and output ports on its right; dragging a cable from an output to an input establishes a dependency. A global clock advances the graph in topological order, and on each tick every node reads its inputs, updates its internal state, and renders a small visualization matched to the quantity it computes. A complete analysis is thus laid out left to right as a visible chain (source  $\rightarrow$  transform  $\rightarrow$  statistic  $\rightarrow$  display), and every intermediate value is observable as it changes (Figure 1). This interaction model is a deliberate descendant of the dataflow and analog-computing traditions (visual patch-and-cable environments and the differential analyzers that preceded digital computation), in which the wiring *is* the program.

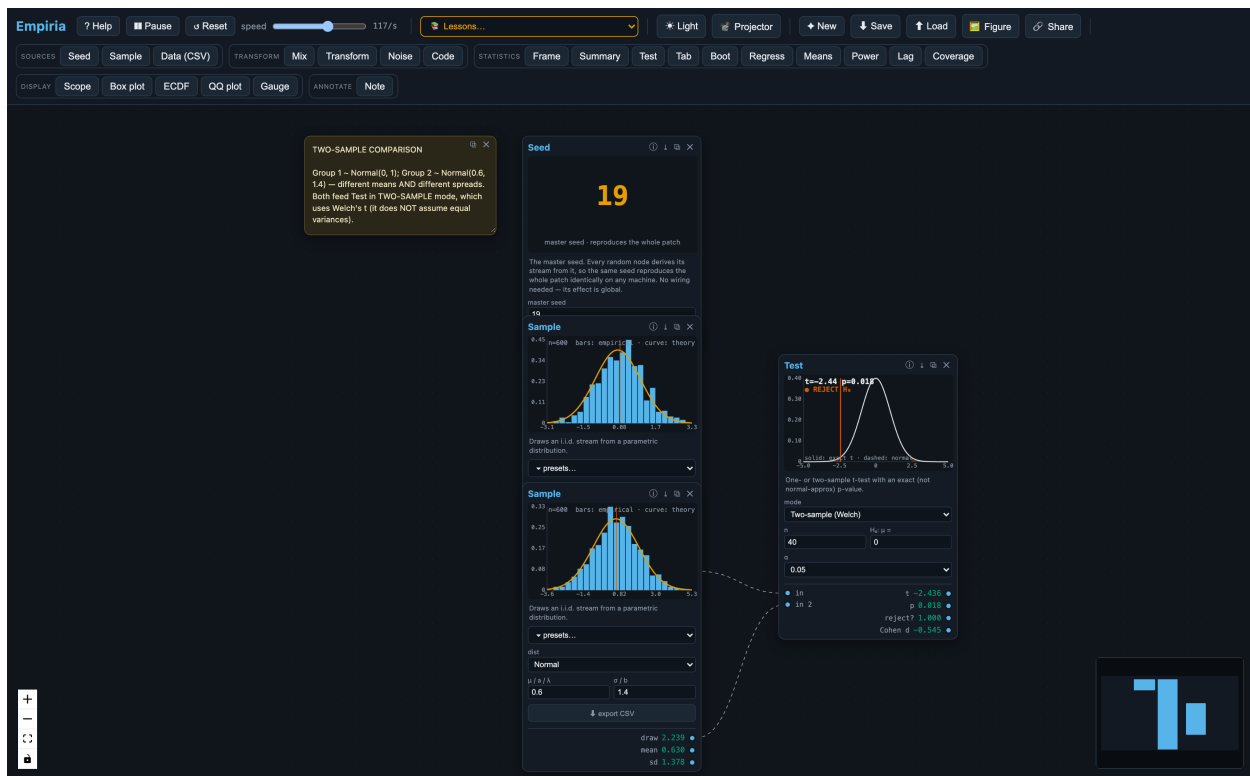


Figure 1. A two-sample comparison assembled on the Empiria canvas. Two independent samplers, each drawing from its own population and showing a live histogram, feed a single test node that overlays the  $t$  reference distribution and reports the statistic and decision. The analysis is *built* as wired modules rather than invoked as a function call, so the data-generating process  $\rightarrow$  sampling  $\rightarrow$  estimation  $\rightarrow$  inference workflow is laid out left to right with every intermediate quantity visible at once.

The clock speed is adjustable from roughly one tick per second (slow enough to watch individual draws accumulate) to several thousand per second for rapid convergence, with the current rate displayed. Light and dark themes, a large-type “projector” mode for lecture rooms, and a colorblind-safe palette support classroom use.

## The engine and reproducibility model

Empiria’s computational core is implemented as a dependency-free TypeScript engine that is independent of the user interface. All randomness derives from a single seeded Mersenne-Twister generator (Matsumoto & Nishimura, 1998); the standard `Math.random` is never used, because it is neither seedable nor reproducible across browsers. Each stochastic node derives its own stream deterministically from the master seed and its node identifier, so that adding or removing a node does not perturb the others. A patch (the complete description of the nodes, their parameters, the wiring, and the master seed) serializes to a small JSON document that can be saved as a file or encoded into a shareable URL; loading it reconstructs the simulation byte-for-byte on any machine. Computation is performed in IEEE-754 double precision with no platform-specific code paths, so a worked example is exactly reproducible across operating

systems and devices.

## Computational footprint

Empiria runs entirely in the browser. There is no server, and after the initial download (a static bundle of roughly 0.4 MB, about 130 KB gzipped) it runs offline. The numerical workload is light: a typical patch ticks well above the interactive clock cap of a few thousand updates per second, so on the low-power hardware typical of classrooms (Chromebooks, tablets) the on-screen frame rate, not the arithmetic, is the limiting factor. Full benchmark figures and a reproducible script are in the Supplementary Materials (Appendix B).

## The module library

The modules are organized on the palette into five families (Table 1). Sources generate or import data; transforms reshape a signal; statistics estimate, test, or summarize; displays render a signal; and an annotation node lets the instructor leave explanatory text that travels with the patch.

Table 1. *The Empiria module library.*

| Family     | Modules                                                                                                                                                                                                                                                                              |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sources    | Seed (master seed), Sample (parametric distribution with real-world presets), Data (CSV import)                                                                                                                                                                                      |
| Transform  | Mix (mixtures), Transform ( $y = a \cdot f(x) + b$ ), Noise (measurement error), Code (continuous $\rightarrow$ ordinal/Likert)                                                                                                                                                      |
| Statistics | Frame (mean/SD/SE), Summary (descriptives), Test (one/two-sample $t$ ), Tab ( $\chi^2$ and Cramér's V), Boot (BCa bootstrap), Regress (OLS), Means (sampling distribution of the mean), Power (power / Type-I error), Lag (autocorrelation), Coverage (confidence-interval coverage) |
| Display    | Scope (time trace), Box (box plot), ECDF, QQ (normal quantile–quantile), Gauge (units readout)                                                                                                                                                                                       |
| Annotate   | Note (editable, saved with the patch)                                                                                                                                                                                                                                                |

Each statistic ships with a visualization designed for the concept it represents: a histogram with a confidence band, a scatterplot with a fitted line and prediction band, a null  $t$ -distribution with shaded rejection regions and the normal approximation overlaid for contrast, a bootstrap distribution annotated with its bias-correction diagnostics, and a confidence-interval “coverage” display that stacks repeated intervals and colors those that miss the true value.

## Pedagogical rationale

Empiria’s design draws on established findings from the statistics-education and mathematical-cognition literatures, which it operationalizes through the principles below.

Dynamic, manipulable visualization supports the construction of sampling intuitions. Students who manipulate parameters and watch sampling distributions respond in real time develop more durable inferential intuitions than those who encounter the same material as static figures (Chance et al., 2004; delMas et al., 2007; Garfield & Ben-Zvi, 2009; for a recent review, Gok & Goldstone, 2024). Empiria adopts this throughout: every parameter is a control, every estimator is a live trace, and every confidence interval is a band that visibly widens and narrows with the sample.

Concrete-to-abstract sequencing scaffolds inferential thinking. Empiria’s patch grammar lets a procedure be *enacted* before it is *symbolized*. A two-sample  $t$ -test is, in Empiria, a literal sequence of objects the student wires together (two samplers, a sampling-window estimator on each branch, and a test module that consumes both), so that when the algebraic formula is later introduced, each symbol corresponds to a module the student has already manipulated.

Live notation ties manipulation to symbolic form. A persistent difficulty in statistics is that formulas stay inert: a student can read  $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$  without connecting any symbol to something they can change. Empiria can render, beneath every node, the formula that node computes in standard mathematical notation, written as a chained equality (the symbolic expression, the current values substituted into it, and the result) that updates on every tick (Figure 2). Manipulating the patch is therefore reflected immediately in the equation: as a sample grows the student watches  $s$  shrink and the denominator  $s/\sqrt{n}$  shrink with it; widening the gap  $\bar{x} - \mu_0$  drives the  $t$ -statistic up; a Transform node shows  $y = 2 \cdot x + 1$  evaluating a specific input to a specific output beside the scatter it produces. The same patch can thus be read two ways at once: picture-first for intuition, and notation-first for formalism. The symbolic layer becomes a manipulable object rather than a static artifact, which directly addresses the symbol-grounding gap the concrete-to-abstract literature identifies. Because the notation is rendered with the browser’s native MathML, the feature adds no dependency and no meaningful weight.

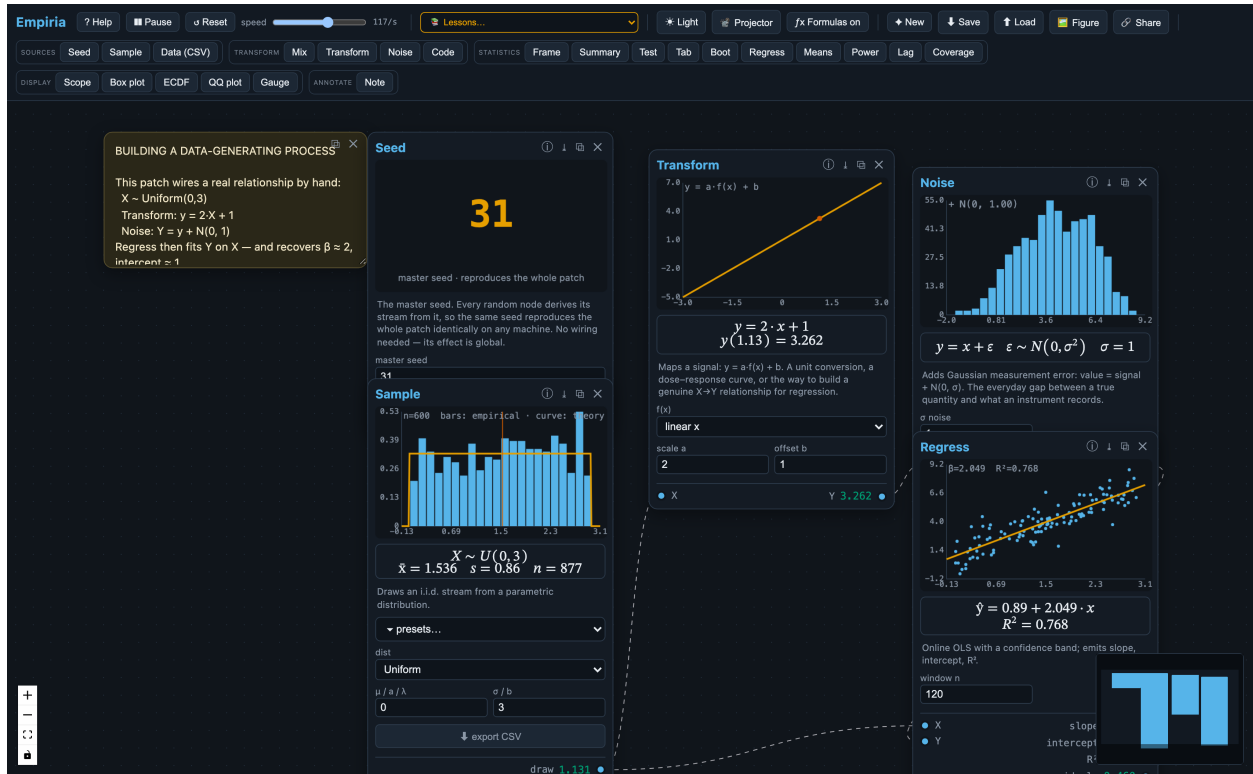


Figure 2. The live-notation view (toggled on). Beneath each node, the formula it computes is rendered in standard notation with the current values substituted in: the Sample node shows its data-generating distribution and running estimates, the Transform node shows  $y = 2 \cdot x + 1$  evaluated at a specific input, the Noise node shows  $y = x + \epsilon$  with  $\epsilon \sim N(0, \sigma^2)$ , and the Regress node reports the recovered  $\hat{y} = a + b \cdot x$  and  $R^2$ . As the user changes a parameter or rewires the graph, every equation updates on the next tick.

Composition makes the inferential workflow visible. Because the output of any module is routable into any other, the data-generating process  $\rightarrow$  sampling  $\rightarrow$  estimation  $\rightarrow$  inference  $\rightarrow$  diagnostics pipeline is a single wired chain rather than a sequence of disconnected applets. This directly targets the structure the reform literature identifies as the learning objective (Cobb, 2007; Tintle et al., 2015).

Modular composition invites customization and independent exploration. Because activities are assembled from interchangeable parts rather than delivered as fixed applets, a student is not confined to a scripted task: a patch can be rewired, one distribution or estimator swapped for another, a parameter nudged so that the downstream consequences propagate visibly, or something the instructor never specified built from scratch (Figure 3). This open-ended, constructionist mode of understanding by building and tinkering (Papert, 1980) is supported directly by the dataflow canvas and by the analog-computing intuition it inherits: a model is something one assembles from manipulable parts and operates, not a result one reads off. The visual immediacy of the per-node displays makes such exploration legible, because a change made anywhere is immediately visible everywhere downstream.

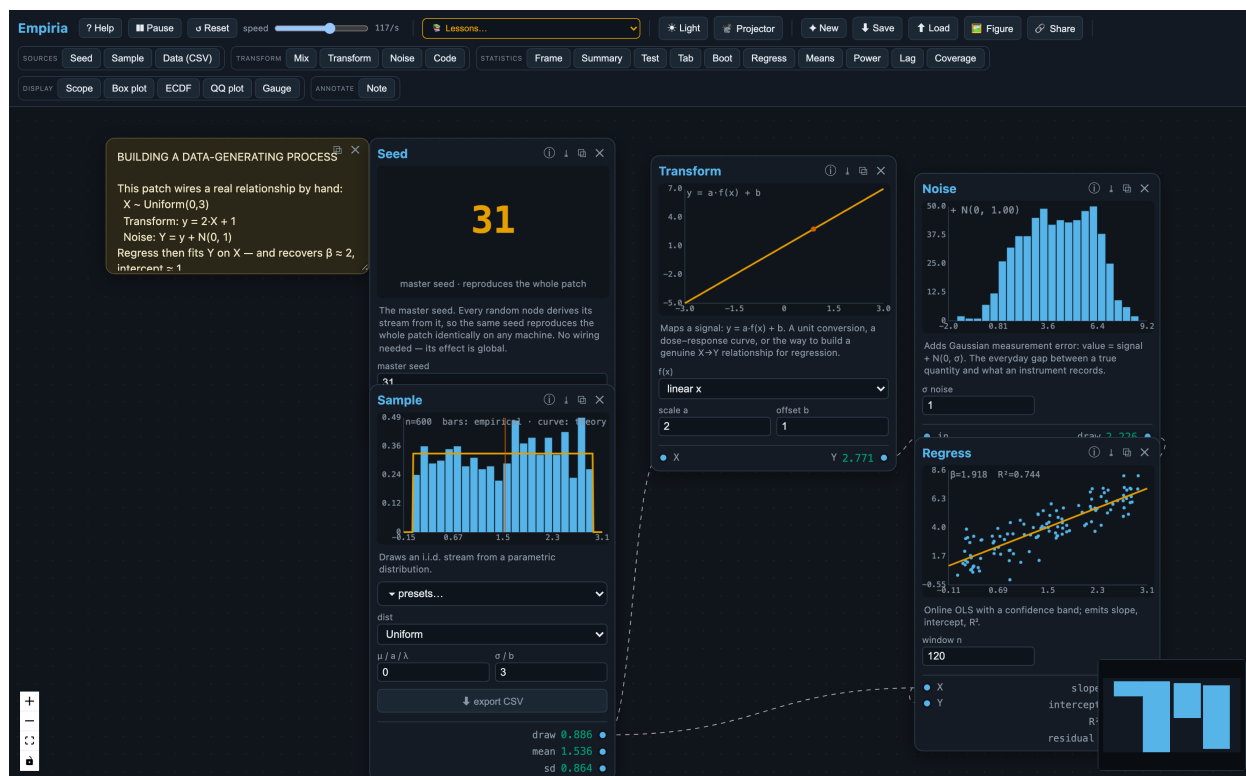


Figure 3. Building a data-generating process from interchangeable parts. A uniform predictor (Sample) is passed through a deterministic Transform ( $y = 2x + 1$ ) and a Noise module before a Regress node recovers the relationship online and reports the fitted slope and intercept. Because the model is *assembled* rather than scripted, a student can swap the transform, change the amount of noise, or rewire the graph and watch every downstream view update. This is the build-and-tinker, analog-computing mode of working that the design is meant to invite.

These principles are operationalized in sixteen bundled lessons, each a self-contained worksheet consisting of a patch plus an explanatory note, and a four-step guided tour (Law of Large Numbers  $\rightarrow$  bootstrap  $\rightarrow$   $t$ -test  $\rightarrow$  confidence-interval coverage) with previous/next navigation. The same modules serve a range of audiences: a general-education class can place a sampler and a sampling-window estimator and watch the standard error shrink as  $1/\sqrt{n}$ ; an introductory course can add regression, testing, and the bootstrap as a live picture of the workflow students will later carry out in R; and a graduate methods seminar can use the platform to interrogate small-sample behavior or to compare bootstrap intervals against analytic ones. What differs across audiences is the depth at which the underlying mathematics is examined, not the tool.

## Teaching and learning context

Empiria is designed for the practising teacher, not the developer. Because it runs in the browser with no installation and ships with ready-to-use lesson patches, the activation cost is low: opening a lesson from the menu and either projecting it on a screen or sharing the link is enough

to start using it.

**Audiences.** The tool serves three broad audiences with the same modules but different depth of treatment. *General-education and first quantitative courses* (a one-semester statistics requirement for non-majors, for example) can use the sampling-distribution, law-of-large-numbers, and coverage activities as concept-building demonstrations without dwelling on the algebra. *Introductory undergraduate statistics* across the social and life sciences builds the inferential chain (sampling, estimation,  $t$ -tests, confidence intervals, the bootstrap, regression) with the same modules students will later use in R or Python; the live picture serves as the intuition that the formal treatment then names. *Methods courses for advanced undergraduate and postgraduate students* use the canvas to interrogate small-sample behaviour, Type-I error and power, model misspecification through the Transform  $\rightarrow$  Noise  $\rightarrow$  Regress chain, and the bootstrap's coverage properties.

**Modes of use.** The lessons are short enough to drop into a 50–75-minute class and flexible enough to support several formats:

- *In-lecture demonstration* (5–15 min): the instructor opens a patch on a projector and steers it through the concept; the built-in *projector* mode enlarges the on-panel text for a lecture room.
- *Computer-lab activity* (30–75 min): students each open the same lesson, follow the prompts in the embedded Note, modify a parameter or two, and report back.
- *Pre-class warm-up or homework* (10–30 min): the instructor shares a patch URL; students open it on a laptop, tablet, or Chromebook and bring a screenshot or an answered prompt to class.
- *Group exploration*: small groups assemble a patch from a specification (“build a data-generating process whose regression slope is hidden by the noise”) and compare their solutions.

**Pacing and preparation.** A typical activity runs 10–45 minutes in class; preparation for an instructor familiar with the tool is on the order of ten minutes per lesson, because lesson patches load with explanatory Notes and sensible defaults. No software installation is required on lab machines, and patches saved as shareable links travel as ordinary URLs. The development context of this article is a methods-teaching role in the social sciences (the author is a post-doctoral researcher at the Institute of Communication and Public Policy at USI), but the tool is discipline-agnostic: the same lessons serve any course in which sampling, inference, and regression are introduced.

## Illustrative classroom use

Each bundled lesson loads as a self-contained worksheet: a patch wired on the canvas plus an explanatory Note that states the goal, what to look at, and what to try. The six lessons below show the range; for each we give the typical audience, the mode of use, an approximate duration, what the student does and observes, and the misconception or learning objective it targets.

Sampling-distribution and the law of large numbers (general-education or introductory undergraduate; in-lecture demonstration or pre-class warm-up; 10–15 minutes). A Seed feeds a Sample drawing from a Normal population into a Frame in growing mode, which displays the running mean and standard error. As the clock runs, the mean settles onto the true value and the standard error contracts as  $1/\sqrt{n}$ . Students then lower the population SD or change the seed and re-run: the *shape* of the convergence is the same, the specific path differs. The activity targets the common misconception that one sample’s mean either “is” or “is not” close to the truth, and gives students an embodied feel for sampling variability and the role of  $n$ .

What “95% confidence” really means (introductory undergraduate or methods; lab or homework; 15–25 minutes). A Coverage node repeats the same experiment again and again: each repetition draws a fresh sample, builds a  $t$ -interval for the mean, and asks whether the true value falls inside. The display stacks the intervals and colours those that miss. After a few hundred repetitions the empirical coverage approaches the nominal 95%. The activity gives an operational meaning to the confidence guarantee, a property of the *procedure* rather than of any one interval, and directly confronts the textbook misreading that “there is a 95% chance the true mean is in *this* interval.”

Bootstrap sampling distribution (introductory undergraduate or methods; lab or homework; 20–30 minutes). A Sample drawn from a skewed (exponential) parent feeds a Boot node that resamples with replacement and shows the bootstrap distribution of the chosen statistic, annotated with its bias-corrected and accelerated interval (Figure 4) (Efron, 1987; Hesterberg, 2015). Students change the statistic from mean to median to standard deviation, change  $n$  and  $B$ , and discuss how the bootstrap distribution narrows. The sampling distribution is now something a student constructed and watched form, rather than something a textbook names without showing.

Comparing two groups with Welch’s  $t$  (introductory undergraduate or methods; lab or homework; 25–40 minutes). Two independent Sample nodes draw from Normal populations with potentially different means and spreads; each feeds a histogram and a Test node configured for Welch’s two-sample  $t$  (Figure 1). Students manipulate the group means and spreads and watch the  $t$  statistic, the  $p$ -value, and the reject/retain decision update. The activity makes the link between effect size, sample size, and decision concrete, and sets up a follow-up discussion of false-positive and false-negative risks.

Building a data-generating process from interchangeable parts (methods seminar or advanced undergraduate; lab; 30–45 minutes). A uniform Sample is passed through a Transform applying  $y = a \cdot x + b$ , then through a Noise module that adds normal measurement error, and finally into a Regress node that fits OLS online (Figure 3). Students vary  $a$ ,  $b$ , and  $\sigma$  and watch the recovered slope and intercept hover near the truth while  $R^2$  rises and falls. They then swap the Transform for a square or a sigmoid and discuss how a linear model behaves when the generating process is not, providing an early, visual introduction to the modelling assumptions students will later read about formally.

False positives when nothing is going on (methods seminar or postgraduate; lab or homework; 15–25 minutes). Two identical Sample nodes feed a Power node that repeatedly runs a Welch

$t$  and tallies the rejection rate. With both populations identical, the rejection rate sits near  $\alpha$ . Students then change  $\alpha$  from 0.01 to 0.10 and watch the false-positive rate track it. The activity makes the meaning of “Type-I error rate” something watched rather than asserted, and provides natural setup for a later session on multiple-testing corrections.

In each case the patch is seeded, so an instructor can distribute it, have students run it, and be confident that every result reproduces across machines. Ten further lessons in the same format cover Likert-coding, a mixture of two distributions, the central limit theorem from a skewed parent, QQ plots, exploratory data analysis on an imported CSV, and others; the full list is in the Supplementary Materials (Appendix F).

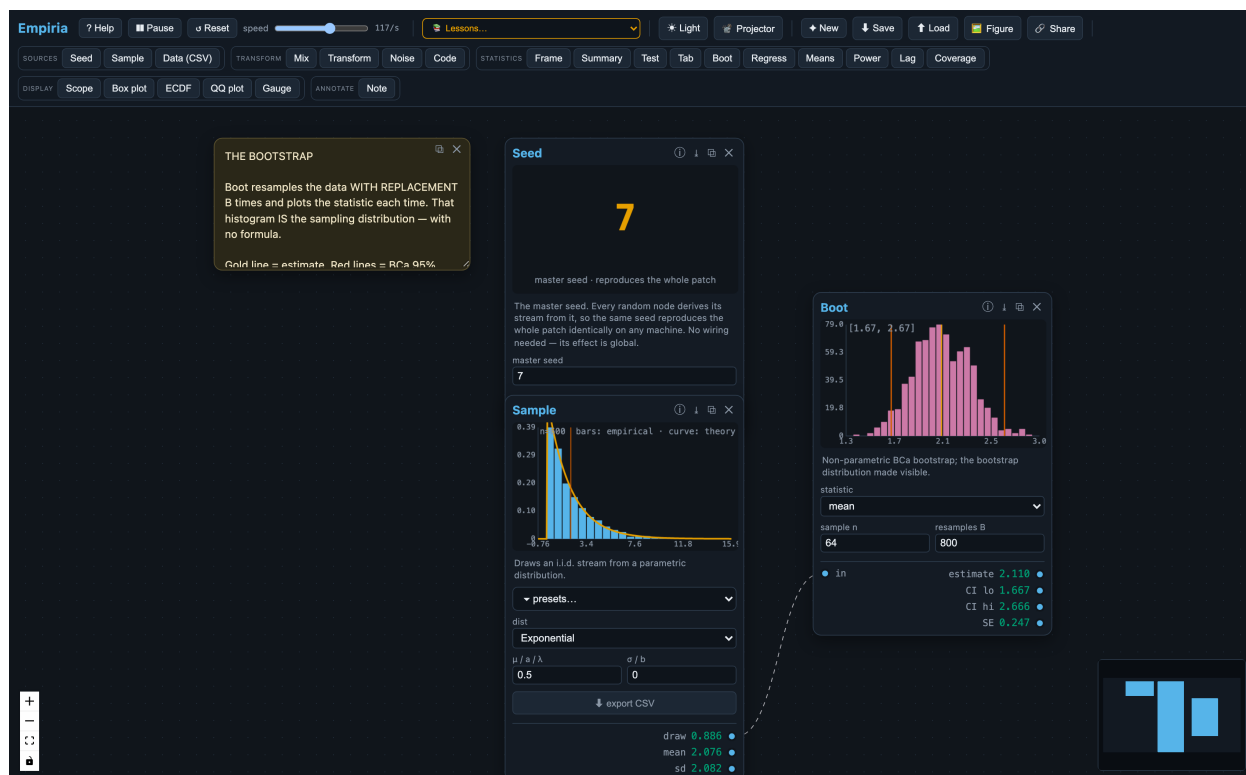


Figure 4. Simulation made visible. A sample drawn from a skewed (exponential) parent is resampled with replacement by the Boot module, whose histogram (magenta) is the bootstrap sampling distribution of the mean, annotated with a bias-corrected and accelerated interval. The sampling distribution is *constructed* empirically and watched as it forms, before any closed-form expression is introduced.

## Informal feedback and design evolution

The design has been shaped by informal practitioner input gathered during development. Students who tried the running application, and a small set of colleagues teaching introductory statistics who responded to an open call for impressions on the tool, raised two recurring themes that influenced both the manuscript and the tool itself.

The first theme was *applicability*: how the tool maps onto a specific course’s teaching goals,

where the example use cases sit, and how an instructor would weave a lesson into existing teaching. In response, the manuscript was rewritten to foreground a *Teaching and learning context* section (audiences, modes of use, pacing, preparation) and to expand *Illustrative classroom use* into the six detailed lesson walkthroughs above, each stating its intended audience, the kind of session it suits, an approximate duration, and the misconception or learning objective it targets.

The second theme was the *activation cost* of a node-and-cable interface for first-time users: how to get started, where to look, what to try first. In response, the application gained a short *guided tour* that walks a new user through the Law of Large Numbers, the bootstrap, the  $t$ -test, and confidence-interval coverage in four steps, lowering the cost of engagement for instructors and students encountering the canvas for the first time. Other recurring observations included appreciation for the live visual feedback as an alternative pathway into statistical ideas, and for the live mathematical-notation view (Figure 2) for its balance of mathematical rigor and a playful, manipulable mode of working.

We treat this informal input as *design evidence*: it has shaped the tool and the manuscript, and explaining how is part of the article’s transparency. Because the feedback was solicited as service-improvement input on the tool rather than as human-subjects research, no identifying or quantified data are collected or reported here, and observations are presented in aggregate, thematic form only. We do not present these themes as evidence of learning outcomes; that is a separate question, taken up in *Limitations and future work*.

## Numerical correctness and reproducibility

### Verifying the numerics

Because the pedagogy depends on the displayed numbers being correct rather than merely plausible, Empiria treats verification as part of the artifact. Where introductory tooling has often relied on normal-theory approximations, Empiria computes closed-form quantities with their exact numerical recipes: the two-tailed Student- $t$  and  $\chi^2$  tail probabilities are evaluated through the regularized incomplete beta and gamma functions using continued-fraction expansions (Press et al., 2007); confidence-interval critical values use the exact  $t$  quantile rather than the  $z \approx 1.96$  approximation; and bootstrap intervals are bias-corrected and accelerated (Efron, 1987).

These routines are checked by an automated test suite that compares each result against R to machine precision, covering the incomplete beta and gamma functions, Student- $t$  probabilities and critical values, the normal CDF and quantile, one-sample and Welch two-sample  $t$ -tests, ordinary least squares, the contingency-table  $\chi^2$  and Cramér’s V, autocorrelation, and the descriptive moments. The Mersenne-Twister generator is checked against its canonical test vector. The suite also verifies system-level properties: byte-identical reproduction under a fixed seed, the contraction of the standard error under the Law of Large Numbers, approximately nominal (95%) confidence-interval coverage, and a Type-I error rate near  $\alpha$  under the null. An accompanying R script regenerates every reference value independently, so a reviewer

can confirm the reference values themselves rather than only the tool’s self-consistency, and a verification document maps each routine to its algorithm, citation, reference command, and test. Finally, because a patch is seeded JSON, a student can export any node’s data to CSV and reproduce the statistic in R as an external check; the same patch and seed yield an identical export on any machine.

## Development with AI coding assistance

Empiria’s implementation was carried out with the help of AI coding assistants (large language models used as programming aids). We disclose this as a matter of scholarly practice, and because it bears on a legitimate concern: software produced with substantial machine assistance may contain plausible-looking but incorrect numerical behavior, and a reader is right to ask whether the displayed statistics can be trusted. The answer here is methodological rather than rhetorical. The trustworthiness of Empiria’s outputs does not rest on the provenance of the code that produces them; it rests on an external verification regime that is independent of how the code was written.

Concretely, every numerical routine is checked against an authoritative reference implementation: R’s statistical functions (`pt`, `qt`, `pchisq`, `pnorm`, `qnorm`, `t.test`, `lm`, `chisq.test`, `acf`, `quantile`), the `boot` package for the BCa interval, and the canonical Mersenne-Twister test vector. Agreement is to machine precision, with Monte-Carlo tolerances stated explicitly where resampling is involved. Those reference values are produced by an independent R script that a reviewer can rerun and compare against the literals asserted in the test suite, so the check confirms the reference values themselves rather than only the tool’s internal self-consistency. A test that compares to R cannot be satisfied by code that is merely persuasive: it passes only if the number is right, whoever or whatever wrote the code.

We also separate authorship from correctness. The statistical methods, the choice of exact recipes over normal-theory approximations, the pedagogical claims, and the interpretation of every result were specified, reviewed, and are vouched for by the author; the AI tools accelerated implementation, they did not make statistical decisions, and the author takes full responsibility for the software and the manuscript. Because the full audit trail maps each routine to its algorithm, citation, reference command, and test, and because the entire suite runs from three commands, any reviewer can reproduce the verification in minutes rather than taking our word for it.

## Data and code availability

Empiria is released under the GPL-3.0 license and runs in any modern browser with no installation. The live application is at <https://kevinschoenholzer.com/empiria/>; the source code, the automated test suite, the verification materials, the bundled lessons, and the documentation are in the public repository at <https://github.com/kevisc/empiria>. The exact version described here is tagged as v2.1.0 in the repository and archived on Zenodo (DOI: <https://doi.org/10.5281/zenodo.20413213>), so that the reviewed software remains citable independently

of later development. The figures in this article are screenshots of the running application; the patches that generate them are included among the bundled lessons.

## Limitations and future work

Empiria’s contribution is a design: an interaction model grounded in established findings about manipulable visualization and simulation-based reasoning, paired with auditable numerics and reproducible, shareable patches. Its defining affordances are visual, analog-computing-inspired, and modular, and they are intended to support guided activities as well as the open-ended, learner-directed exploration in which a student customizes a patch, substitutes one component for another, and reaches a result by building rather than by following a script.

We should be explicit about what this article does not establish. It offers no evidence of learning gains from Empiria itself: the pedagogical claims above are design rationale, grounded in prior findings about related tools, and whether they translate into measurable outcomes is an open empirical question. Reviews of simulation tools are a useful corrective here: interactive visualization does not by itself guarantee understanding: students can watch a sampling distribution narrow and still misread it as a distribution of individual values. The surrounding tasks and design details do much of the work (Gok & Goldstone, 2024). The affordances are available to instructors and students today, but their educational value remains to be demonstrated.

A formal evaluation of learning outcomes is a possible future extension, but we have not undertaken one. Empiria has not been studied in a classroom or experimental setting, and we therefore make no claim about its effect on student understanding, retention, or transfer. What we can claim, and what we have tried to ensure, is that the tool has been shaped toward concrete learning goals and the practitioner needs articulated during its development (reported above in *Informal feedback and design evolution*), and that its numerical outputs are correct and independently verifiable: closed-form routines checked against R to machine precision, the bootstrap validated by its coverage behavior, and an automated test suite that any reviewer can rerun in minutes (full detail in *Numerical correctness and reproducibility* and the Supplementary Materials). What an instructor needs in order to trust the numbers Empiria displays is therefore on offer; what would be needed to claim that an Empiria-augmented lesson moves a learning-outcome measure is a controlled study, which we leave to future work.

Two practical considerations are worth noting for adopters. Because a node-and-cable interface is initially unfamiliar, the bundled guided tour and lesson patches are designed to lower the activation cost and to give students a scaffold from which to begin tinkering; and because the modules are visually engaging, the suggested classroom integration is additive to, rather than a replacement for, a standard analytic workflow. On the technical side, confidence intervals for proportions currently use the normal (Wald) form, and the bootstrap is validated by its coverage behavior and by exact tests of its components rather than by a value-for-value match to a specific resampling implementation.

## Conclusion

Empiria operationalizes the central recommendation of the statistics-education reform literature (that students build the sampling distribution before they name it) by making the inferential workflow a composable, manipulable object rather than a sequence of disconnected applets. By pairing that interaction model with exact, independently auditable numerics and reproducible, shareable patches, it aims to be trustworthy as well as legible: a tool a student can learn from and an instructor can verify.

## Disclosure statement

The author reports no competing interests. No funding supported this work. The informal practitioner feedback referenced in this article was solicited as service-improvement input on the tool rather than as human-subjects research; no identifying information was collected, and observations are reported in aggregate, thematic form only. AI-based coding assistants (large language models) were used as programming aids during development of the software. All statistical methods and numerical results were specified by the author and verified against independent reference implementations, as described in *Numerical correctness and reproducibility*; the author takes full responsibility for the content of the software and the manuscript.

## References

1. American Statistical Association, Guidelines for Assessment and Instruction in Statistics Education (GAISE) College Report 2016, *American Statistical Association*, 2016. <https://www.amstat.org/education/gaise>
2. B. Chance, R. delMas, J. Garfield, Reasoning about sampling distributions, in: D. Ben-Zvi, J. Garfield (Eds.), *The Challenge of Developing Statistical Literacy, Reasoning, and Thinking*, Springer, 2004, pp. 295–323. [https://doi.org/10.1007/1-4020-2278-6\\_13](https://doi.org/10.1007/1-4020-2278-6_13)
3. G. W. Cobb, The introductory statistics course: A Ptolemaic curriculum?, *Technology Innovations in Statistics Education*, 1(2007). <https://doi.org/10.5070/T511000028>
4. R. delMas, J. Garfield, A. Ooms, B. Chance, Assessing students' conceptual understanding after a first course in statistics, *Statistics Education Research Journal*, 6(2007), 28–58. <https://doi.org/10.52041/serj.v6i2.483>
5. B. Efron, Better bootstrap confidence intervals, *Journal of the American Statistical Association*, 82(1987), 171–185. <https://doi.org/10.1080/01621459.1987.10478410>
6. W. Finzer, The data science education dilemma, *Technology Innovations in Statistics Education*, 7(2013). <https://doi.org/10.5070/T572013891>
7. J. Garfield, D. Ben-Zvi, Helping students develop statistical reasoning: Implementing a statistical reasoning learning environment, *Teaching Statistics*, 31(2009), 72–77. <https://doi.org/10.1111/j.1467-9639.2009.00363.x>

8. S. Gok, R. L. Goldstone, How do students reason about statistical sampling with computer simulations? An integrative review from a grounded cognition perspective, *Cognitive Research: Principles and Implications*, 9(2024), 33. <https://doi.org/10.1186/s41235-024-00561-x>
9. T. C. Hesterberg, What teachers should know about the bootstrap: Resampling in the undergraduate statistics curriculum, *The American Statistician*, 69(2015), 371–386. <https://doi.org/10.1080/00031305.2015.1089789>
10. C. Konold, R. Lehrer, Technology and mathematics education: An essay in honor of Jim Kaput, in: L. D. English (Ed.), *Handbook of International Research in Mathematics Education*, 2nd ed., Routledge, 2008.
11. R. H. Lock, P. F. Lock, K. Lock Morgan, E. F. Lock, D. F. Lock, *Statistics: Unlocking the Power of Data*, 3rd ed., Wiley, 2021. StatKey: <https://www.lock5stat.com/statkey>.
12. M. Matsumoto, T. Nishimura, Mersenne Twister: A 623-dimensionally equidistributed uniform pseudo-random number generator, *ACM Transactions on Modeling and Computer Simulation*, 8(1998), 3–30. <https://doi.org/10.1145/272991.272995>
13. S. Papert, *Mindstorms: Children, Computers, and Powerful Ideas*, Basic Books, 1980.
14. W. H. Press, S. A. Teukolsky, W. T. Vetterling, B. P. Flannery, *Numerical Recipes: The Art of Scientific Computing*, 3rd ed., Cambridge University Press, 2007.
15. A. Rossman, B. Chance, Using simulation-based inference for learning introductory statistics, *WIREs Computational Statistics*, 6(2014), 211–221. <https://doi.org/10.1002/wics.1302>
16. N. Tintle, B. Chance, G. Cobb, S. Roy, T. Swanson, J. VanderStoep, Combating anti-statistical thinking using simulation-based methods throughout the undergraduate curriculum, *The American Statistician*, 69(2015), 362–370. <https://doi.org/10.1080/00031305.2015.1081619>
17. U. Wilensky, *NetLogo*, Center for Connected Learning and Computer-Based Modeling, Northwestern University, 1999. <http://ccl.northwestern.edu/netlogo/>